

Statistical Risk Modeling in R

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Statistical Risk Modeling in R

Executive Summary

This public-safe project models the probability that a synthetic B2B account will require escalation review in the next quarter. The selected interpretable GLM is **Full operating model**, chosen using repeated stratified cross-validation and checked on a holdout set.

On the holdout set, the model achieved AUC **0.719**, log loss **0.422**, and Brier score **0.129**. Cross-validation produced log loss **0.366** and AUC **0.797** for the selected model.

At a 20% review threshold, the workflow flags 77 holdout accounts (27.5%), captures 28 of 48 observed escalations, and produces PPV 36.4%. This threshold is a planning option, not a final policy: review capacity and intervention cost should set the operating point.

The ranking view is stronger than a threshold alone: the highest-risk decile has 2.71x lift over the base event rate, and the top two deciles capture 50.0% of observed escalations. Under the illustrative economics assumptions documented below, the best tested threshold is 15% with net value \$127,350.

Data Overview

The dataset contains 1,400 synthetic account records. The generated escalation rate is 17.1%, and 10.8% of accounts have missing training-completion values.

Predictors include account segment, region, contract value, tenure, product usage, active-seat ratio, training completion, support tickets, response time, prior incidents, and implementation complexity. No real customer, student, patient, credential, or private course data is used.

Model Journey

Candidate logistic models were compared with repeated stratified 5-fold cross-validation on the training split. Log loss is the primary criterion because the business problem needs calibrated probabilities, not only rank ordering. A spline benchmark is included as a flexible model-family check, but the operating model is selected from interpretable GLM candidates.

Model	Selected	Role	Parameters	CV_log_loss	CV_log_loss	CV AUC	Holdout_log_loss	Holdout AUC	Delta CV_log_loss
Full operating model	Yes	Selection candidate	15	0.366	0.005	0.797	0.422	0.719	0.000
Full model with interaction		Selection candidate	16	0.367	0.005	0.798	0.422	0.721	0.000

Model	Selected	Role	Parameters	CV_log_loss	CV_log_loss	CV AUC	Holdout_log_loss	Holdout_AUC	Delta_CV_log_loss
Spline operating benchmark		Benchmark	21	0.371	0.005	0.792	0.424	0.717	0.004
Support load		Selection candidate	9	0.377	0.003	0.781	0.442	0.674	0.011
Usage behavior		Selection candidate	10	0.395	0.005	0.756	0.435	0.686	0.028
Baseline exposure		Selection candidate	6	0.447	0.003	0.619	0.456	0.576	0.081

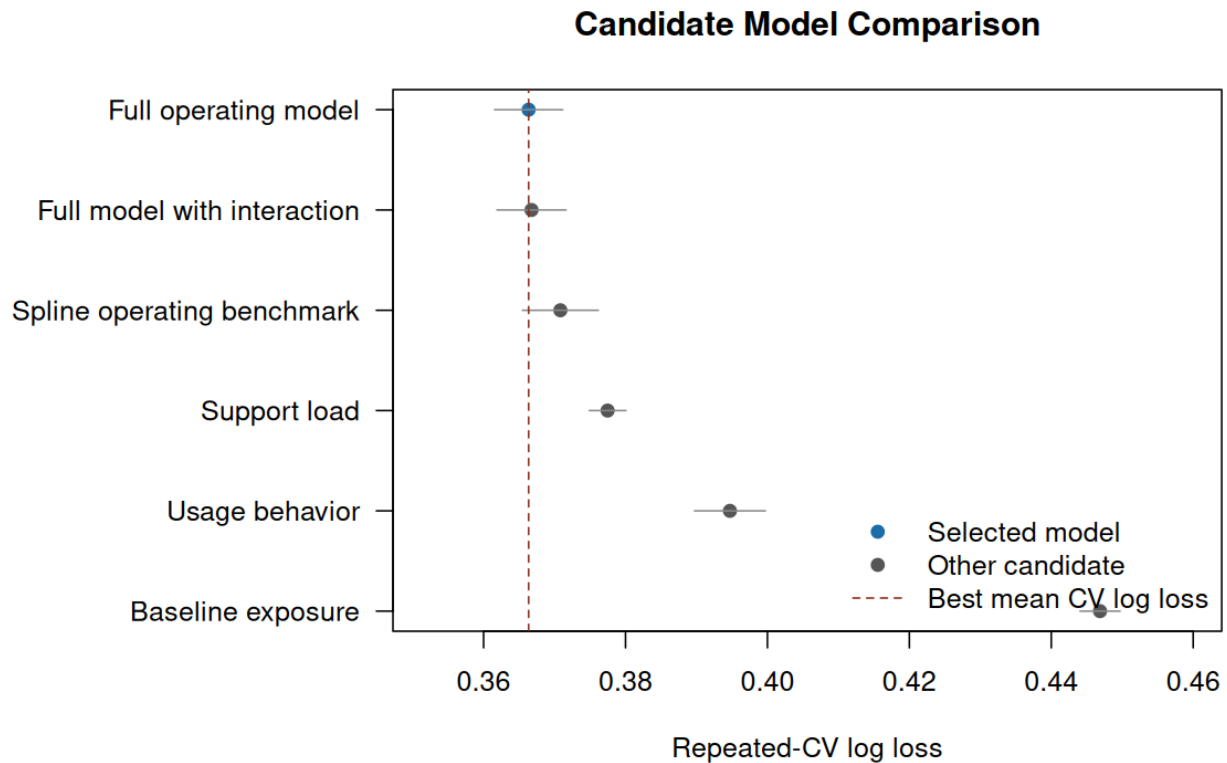


Figure 1: Candidate model comparison

The selected model uses a one-standard-error rule: pick the simplest model whose repeated-CV log loss is statistically close to the best candidate. This keeps the model easier to explain when extra terms do not materially improve probability quality.

Validation Metrics

Metric	Value
Selected model	Full operating model

Metric	Value
Selection rule	Simplest model within one standard error of best repeated-CV log loss
Training rows	1120
Holdout rows	280
Training event rate	17.1%
Holdout event rate	17.1%
Repeated CV folds	5
Repeated CV repeats	6
CV log loss	0.366
CV AUC	0.797
Holdout log loss	0.422
Holdout Brier score	0.129
Holdout AUC	0.719
Calibration intercept	0.035
Calibration slope	0.655
Training log loss	0.349
Training AUC	0.820

Bootstrap intervals give a practical uncertainty band around the holdout metrics.

Metric	Estimate	Bootstrap_95_CI
LogLoss	0.422	0.346 to 0.515
Brier	0.129	0.103 to 0.160
AUC	0.719	0.637 to 0.797

Coefficient Interpretation

The table reports adjusted odds ratios for the selected model. Continuous predictors are scaled to business-readable increments where useful.

Predictor	Scale	Odds_ratio	CI_95	P_value
Contract value, log scale	1-unit / level change	0.84	0.57 to 1.26	0.406
Tenure	12-unit change	0.86	0.76 to 0.96	0.007
Segment: Mid-Market vs SMB	1-unit / level change	0.90	0.51 to 1.57	0.701
Segment: Enterprise vs SMB	1-unit / level change	0.57	0.24 to 1.33	0.194
Segment: Strategic vs SMB	1-unit / level change	1.54	0.47 to 5.11	0.476
Implementation complexity: Medium vs Low	1-unit / level change	1.05	0.68 to 1.62	0.823
Implementation complexity: High vs Low	1-unit / level change	1.26	0.74 to 2.15	0.390
Product usage rate	0.1-unit change	0.89	0.75 to 1.06	0.200
Active-seat ratio	0.1-unit change	0.86	0.73 to 1.02	0.077
Training completion	0.1-unit change	0.85	0.73 to 0.99	0.036
Training completion missing	1-unit / level change	0.88	0.49 to 1.57	0.655

Predictor	Scale	Odds_ratio	CI_95	P_value
Support tickets, last 90 days	3-unit change	1.81	1.34 to 2.46	<0.001
Average response time	12-unit change	1.39	1.11 to 1.76	0.005
Prior incident	1-unit / level change	2.19	1.48 to 3.22	<0.001

Diagnostics

ROC checks ranking quality, while calibration checks whether predicted probabilities are on the right scale across risk bands.

Risk_band	N	Mean_predicted	Observed_rate	Expected_events	Observed_events
Band 1	35	2.0%	5.7%	0.7	2
Band 2	35	3.9%	5.7%	1.4	2
Band 3	35	6.1%	17.1%	2.1	6
Band 4	35	8.9%	8.6%	3.1	3
Band 5	35	12.5%	8.6%	4.4	3
Band 6	35	18.1%	14.3%	6.3	5
Band 7	35	28.0%	34.3%	9.8	12
Band 8	35	54.4%	42.9%	19.0	15

Diagnostic	Estimate	Interpretation
Calibration intercept	0.035	Near 0 means predicted risk is not systematically high or low
Calibration slope	0.655	Near 1 means predicted probabilities are not overly extreme or compressed

Segment and implementation-complexity calibration checks help identify where monitoring should continue after deployment.

Subgroup	Level	N	Mean_predicted	Observed_rate	Calibration_gap	Observed_events
segment	Enterprise	58	8.8%	13.8%	5.0%	8
segment	Mid-Market	112	15.6%	14.3%	-1.3%	16
segment	SMB	92	23.0%	22.8%	-0.2%	21
segment	Strategic	18	17.9%	16.7%	-1.2%	3
implementation_high	High complexity	55	35.7%	38.2%	2.5%	21
implementation_low	Low complexity	111	12.3%	9.9%	-2.4%	11
implementation_medium	Medium complexity	114	11.9%	14.0%	2.1%	16

Threshold Interpretation

A threshold turns probabilities into an operating workflow. Lower thresholds catch more events but increase review burden; higher thresholds concentrate risk but miss more events.

Threshold	Flagged	Flagged_share	Events_captured	Sensitivity	Specificity	PPV	NPV
10%	147	52.5%	35 of 48	72.9%	51.7%	23.8%	90.2%
15%	103	36.8%	31 of 48	64.6%	69.0%	30.1%	90.4%
20%	77	27.5%	28 of 48	58.3%	78.9%	36.4%	90.1%

Threshold	Flagged	Flagged_share	Events_captured	Sensitivity	Specificity	PPV	NPV
25%	59	21.1%	24 of 48	50.0%	84.9%	40.7%	89.1%
30%	46	16.4%	19 of 48	39.6%	88.4%	41.3%	87.6%
35%	38	13.6%	16 of 48	33.3%	90.5%	42.1%	86.8%

The table below translates the threshold choices into an illustrative operating economics frame. Assumptions: \$750 review cost per flagged account, \$12,000 avoided escalation cost, and 55% intervention effectiveness for captured events. These are scenario assumptions, not claims about a real business.

Threshold	Flagged	Events_captured	Avoided_loss	Review_cost	Net_value
10%	147	35	\$231,000	\$110,250	\$120,750
15%	103	31	\$204,600	\$77,250	\$127,350
20%	77	28	\$184,800	\$57,750	\$127,050
25%	59	24	\$158,400	\$44,250	\$114,150
30%	46	19	\$125,400	\$34,500	\$90,900
35%	38	16	\$105,600	\$28,500	\$77,100

Lift and Prioritization

A ranked queue is often more useful than a single classification cutoff. The lift table shows how much event concentration appears in each predicted-risk decile.

Decile	N	Mean_predicted	Observed_rate	Events	Lift	Cumulative_capture
1	28	58.4%	46.4%	13	2.71x	27.1%
2	28	32.6%	39.3%	11	2.29x	50.0%
3	28	22.1%	17.9%	5	1.04x	60.4%
4	28	16.2%	10.7%	3	0.62x	66.7%
5	28	12.0%	10.7%	3	0.62x	72.9%
6	28	9.1%	10.7%	3	0.62x	79.2%
7	28	6.9%	17.9%	5	1.04x	89.6%
8	28	4.9%	7.1%	2	0.42x	93.8%
9	28	3.5%	7.1%	2	0.42x	97.9%
10	28	1.8%	3.6%	1	0.21x	100.0%

Risk categories provide an executive-friendly bridge between probabilities and action queues.

Risk_category	Accounts	Share	Mean_predicted	Observed_rate	Observed_events
Monitor	133	47.5%	5.0%	9.8%	13
Watch	70	25.0%	14.2%	10.0%	7
Review	39	13.9%	26.2%	30.8%	12
Priority	38	13.6%	52.9%	42.1%	16

Sensitivity Analysis

The primary model imputes missing training completion with segment medians and includes a missingness indicator. The sensitivity model uses a conservative assumption: missing training completion is treated as zero. This tests whether the operating story depends on a favorable missing-data assumption.

Measure	Primary	Sensitivity
Holdout log loss	0.422	0.421

Measure	Primary	Sensitivity
Holdout Brier score	0.129	0.129
Holdout AUC	0.719	0.719
Mean absolute probability change	Reference	0.16%
Accounts changing risk category	Reference	2 of 280
Maximum absolute probability change	Reference	1.64%

Scenario Profiles

Scenario profiles translate the model into concrete account-review examples with probability intervals. These are synthetic profiles used for communication and model interpretation.

Scenario	Segment	Complexity	Usage	Support_tickets	Response_time	Open_incidents	Predicted_risk	GL_95	Risk_category
Stable renewal	Mid-Market	Low	82.0%	1	7	0	2.0%	1.1% to 3.7%	Monitor
Enablement watchlist	SMB	Medium	55.0%	4	16	0	25.5%	15.5% to 39.0%	Review
Priority intervention	Enterprise	High	35.0%	8	28	1	73.0%	51.2% to 87.4%	Priority

Decision-Support Implications

- Use predicted risk to prioritize human account review, not to automate account decisions.
- Choose the review threshold based on available review capacity, expected intervention cost, and tolerance for missed escalations.
- Monitor calibration by segment and implementation complexity before treating risk bands as stable operating categories.
- Refit and recalibrate the model when product usage, support operations, or customer mix materially changes.

Reproducibility

Rebuild the full evidence packet with:

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make all
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The build uses base R only and does not require package installation, network access, credentials, or private data.

Public-Safety Statement

This report is an original public-safe portfolio artifact. It excludes private coursework prompts, exams, rubrics, syllabi, lecture transcripts, source datasets, personal data, patient data, school-private records, credentials, and copyrighted source documents.

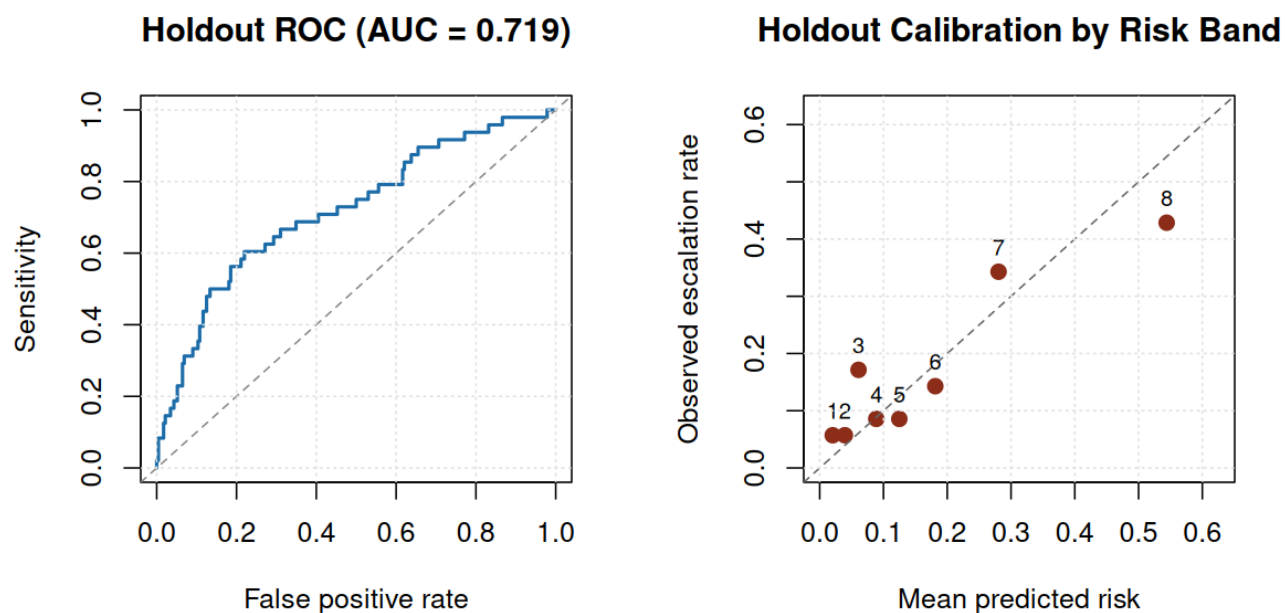


Figure 2: ROC and calibration diagnostics

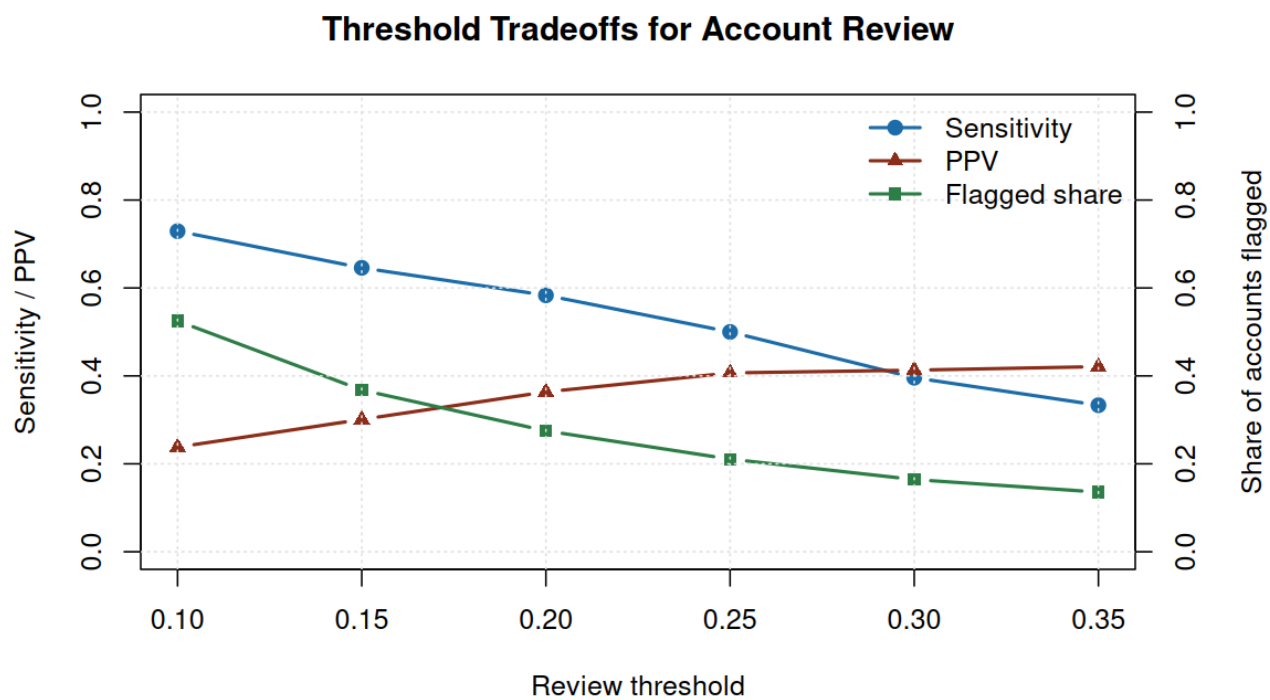


Figure 3: Threshold tradeoffs

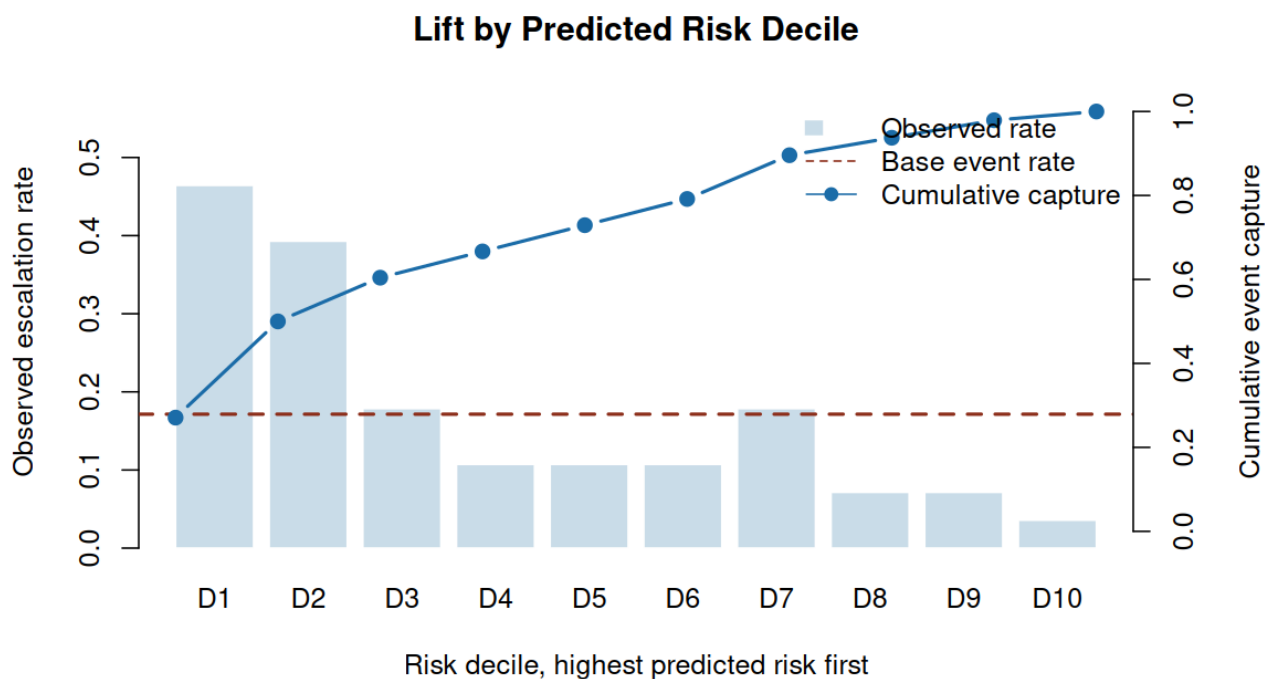


Figure 4: Lift by predicted risk decile

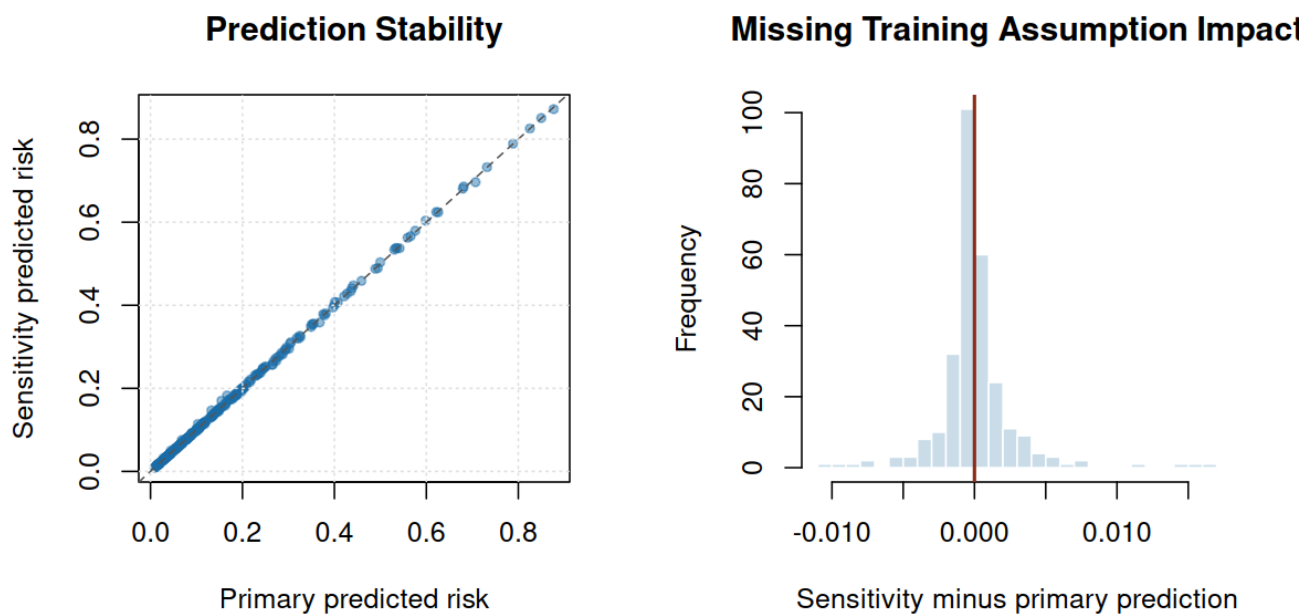


Figure 5: Sensitivity analysis

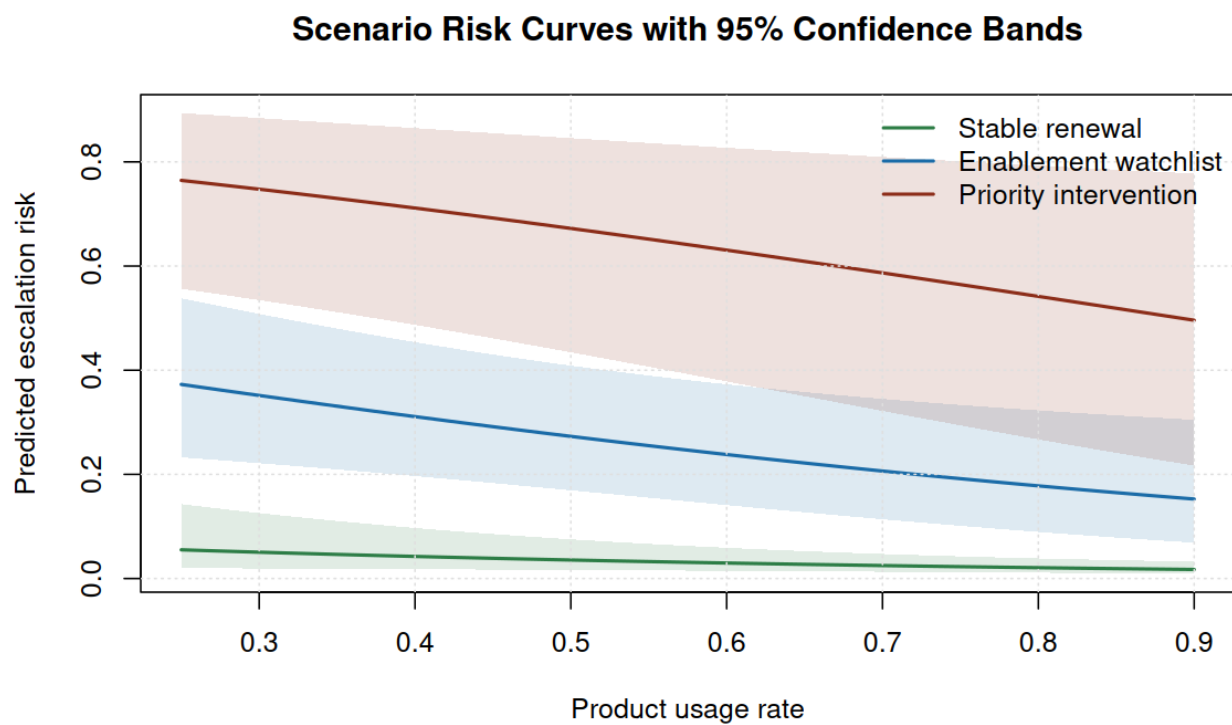


Figure 6: Scenario risk curves